

Ad Omnia

Ad omnia in perpetuum



universidade de aveiro
theoria poiesis praxis



instituto de
telecomunicações

Context & Problem



Data scattered
across multiple
sources



Delay in critical
scenarios.



Severe damage to
infrastructures and
citizens.

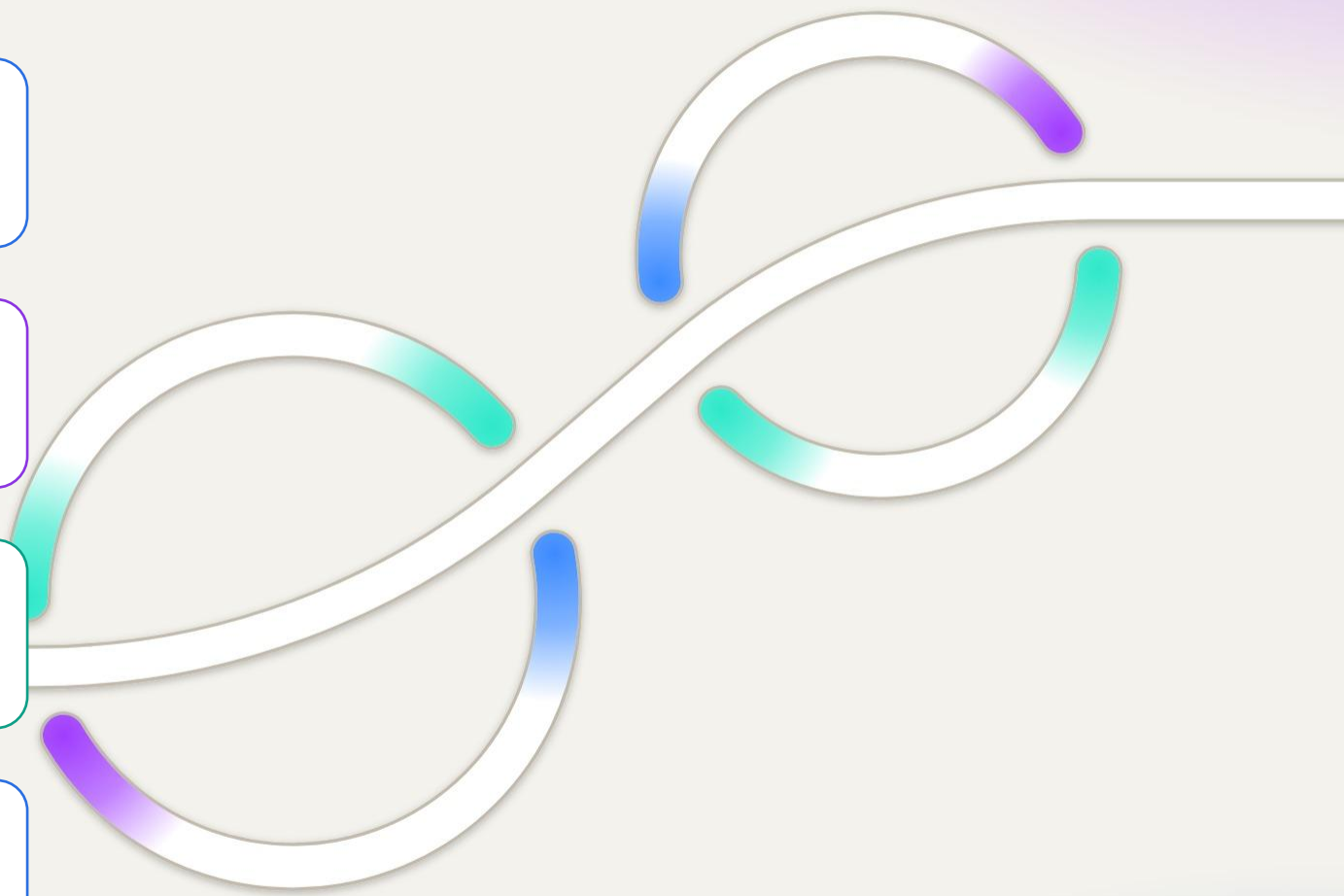
Goals

Integrate diverse data into a single platform

Build a COP (Common Operational Picture) interface for decision support

Develop a predictive AI engine to anticipate risk scenarios

Automated recommendation for resource allocation, routing, etc.



Expected results

Functional COP platform for spatio-temporal data visualization, featuring public and private access tiers

Unified data ingestion engine with built-in connectors for public data sources

Trained Predictive AI model for risk scenario forecasting

Scalable, plugin-based architecture designed for future platform expansion

Tactical recommendation system providing automated suggestions for asset allocation and routing



The team



Bernardo Reis

ML Engineer



Diogo Silva

SCRUM Master



Fernando Santos

Architect



Guilherme Martins

DevOps



Miguel Neto

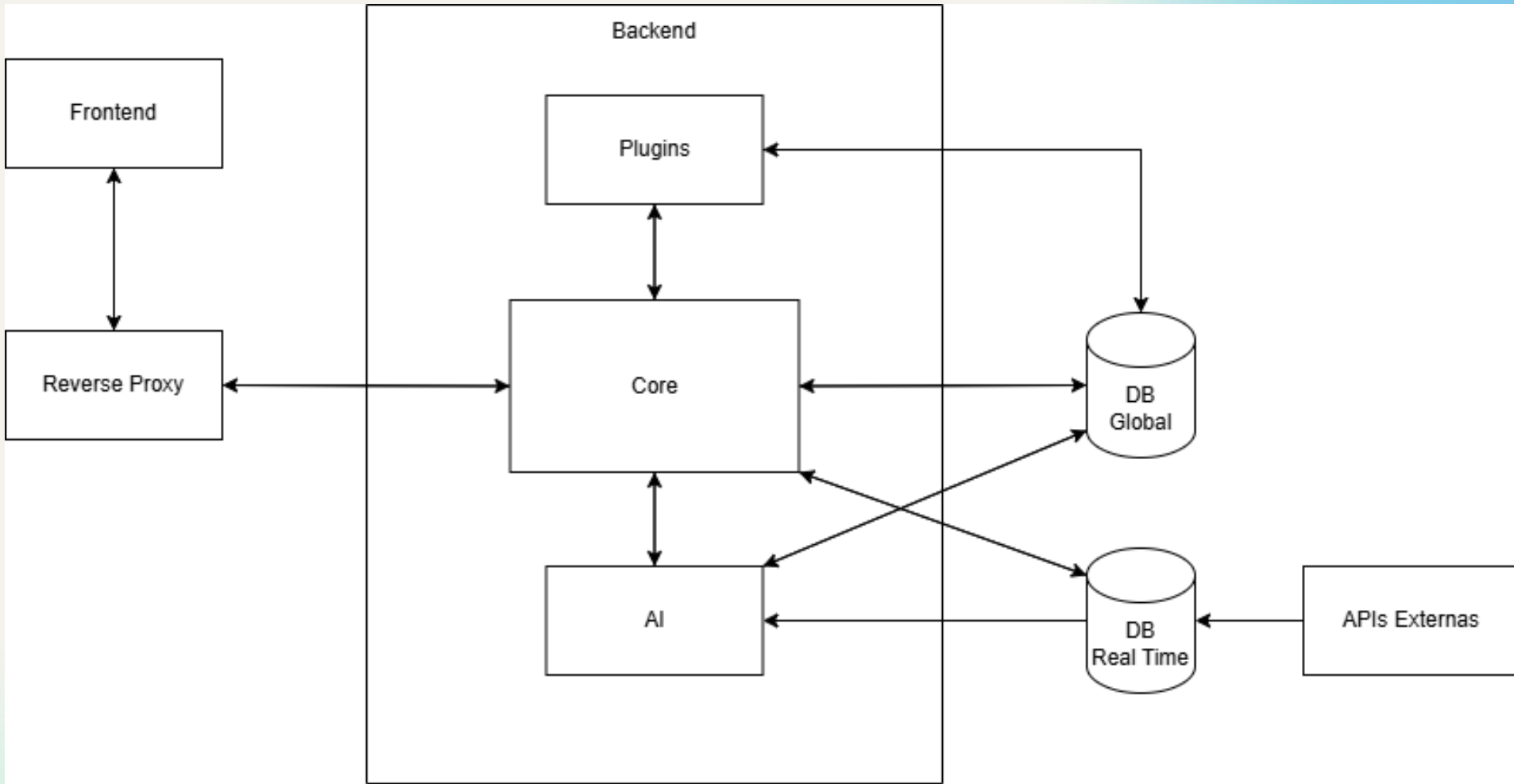
Backend

State of the Art

FEATURE	Palantir Gotham	Thales HexaForce	SADO	HxGN OnCall	Esri ArcGIS	Ours Ad Omnia
Native connectors for Portuguese public data (ANEPC, SNS, Fogos.pt)	☐	☐	☒	☐	☒	☒
Predictive incident progression	☒	☒	☐	☒	☒	☒
Interoperability between police, firefighters and EMS	☒	☐	☒	☒	☒	☒
Automatic resource allocation and routing recommendations	☒	☒	☐	☒	☒	☒
Plugin architecture for new modules and data sources	☒	☒	☐	☒	☒	☒
Dual-use architecture (civil and defense)	☒	☐	☐	☐	☒	☒



High-Level Architecture



Project Calendar

Milestone	Date	Tasks
M1	22/09 - 29/09	• GitHub Organization (Bernardo) • Project backlog using GitHub Projects (Diogo) • Project logo (Fernando) • State of The Art (Guilherme) • Project Website (Miguel, Fernando) • APIs and Data gathering
M2	30/09 - 13/10	• Personas, Requirements and User Stories gathering • Architecture Design • ER and Classes Diagram • High-fidelity UI mockups
M3	14/10 - 03/11	• UI implementation • Core functionalities • Usability tests • Development environment setup • CI/CD pipeline



Project Calendar

Milestone	Date	Tasks
M4	04/11 - 15/12	• MVP features: ○ Unified data ingestion engine ○ COP interface ○ Predictive AI ○ Automated recommendation system • Tests for QA • Stabilization and deployment
M5, ...	15/12 - 01/06/27	• Further development and refining • Commercial video • Poster for Students@DETI • Students@DETI demonstration • Final Report

Communication Plan

1 Website

Includes an overview of the project, as well as documentation and links to our repositories

2 Events

Participation in events: exhibitions and competitions (e.g., Think Tank)

3 Students@DETI

Presentation with demonstration of the project.

4 Other

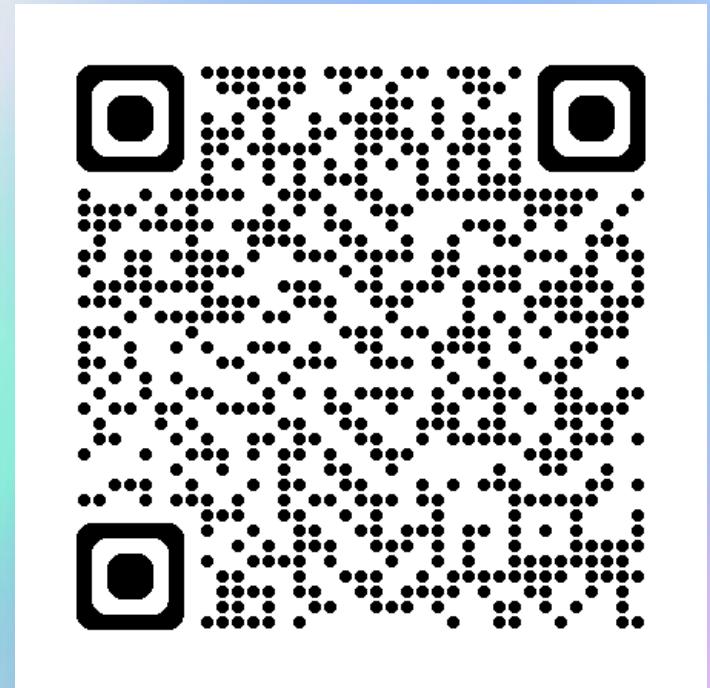
Meetings with emergency and security agencies

Thank you

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GitHub: <https://github.com/Ad-Omnia>

Website: <https://ad-omnia.github.io/>



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